

Amrita Singh

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Summary

Clinical research scientist with a **PhD in neuroscience** and 8+ years of experience across clinical trials and academic research. Currently **leading operations and analysis for a Phase II clinical trial** at Bionic Sight, supporting regulatory submissions and trial design decisions. Skilled in statistical analysis, experiment design, data visualization, and scientific communication.

Education

PhD in Neuroscience

Advisor: Dr. Karel Svoboda

Johns Hopkins University

August 2016 - January 2023

Topic: *In vivo* voltage imaging and transcriptomics in mouse motor cortex | [Thesis](#)

Studied the neural basis of movement, designing **custom analysis tools in Python** to interpret complex, high-dimensional neural activity imaging data. | [List of publications](#)

B. Tech in Biology and Bio-Engineering

Grade: 9.4/10

Indian Institute of Technology, Kanpur

August 2012 - May 2016

Work Experience

Bionic Sight LLC

Clinical Research Scientist

January 2024 - present

- **Lead operations for a Phase II clinical trial** of a combination therapy (optogenetics + external stimulation device) for blindness, ensuring **compliance** with FDA regulations and **alignment** with company objectives.
- Oversaw assessment of patients' visual function across **multiple clinical endpoints**, ensuring data integrity and adherence to protocol.
- Advised the CEO on **data-driven decision-making** including patient enrollment strategies, trial design adjustments, and prioritization of outcome measures.
- **Developed novel data visualizations**; improved **data organization and workflows**; and **prepared figures and written material**; culminating in the company's Regenerative Medicine Advanced Therapy (RMAT) Designation by the FDA in Jan '25.

HHMI's Janelia Research Campus

Postdoctoral Scientist

January - December 2023

- Built **Python-based pipelines** for processing **high dimensional voltage imaging and in situ hybridization data**, and interpreting results.
- Worked with a diverse team including biochemists, experimental neuroscientists and machine learning researchers to develop methods for in vivo voltage imaging.

Skills

Programming

R, Python (NumPy, SciPy, Scikit-learn, TensorFlow), Java

Software/tools

Excel, PowerPoint, Illustrator, Jupyter lab, VS Code

Domain Expertise

Clinical trial design, FDA regulatory processes, endpoint analysis. GCP certified.